

UNIVERSITY OF CAPE TOWN

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II

Advanced Calculus (2AC)

Maximum - Minimum II: One equality constraint

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0.1 Introduction

We will be studying in these notes, problems of the form

$$(P_1) \quad \begin{cases} \text{Find Max/Min of } f(x, y) \\ \text{subject to } g(x, y) = 0. \end{cases}$$

called **optimization problem in $\mathbb{R}^2$ with one equality constraint $g(x, y) = 0$** and also

$$(P_2) \quad \begin{cases} \text{Find Max/Min of } f(x, y, z) \\ \text{subject to } h(x, y, z) = 0 \end{cases}$$

called **optimization problem in $\mathbb{R}^3$ with one equality constraint $h(x, y, z) = 0$** .

We are going to assume that the functions f, g, h are of class $\mathcal{C}^1$, that is, with continuous first order partial derivatives. We are going to study these problems through a well known approach called **the method of Lagrange multipliers**.

So, if we are asked what is the method of Lagrange multipliers for, we can just respond by saying that this is a strategy for finding local maximum or minimum of a function subject to equality constraints. Those constraints are set of equations to be satisfied by the chosen variables.

In these notes, we are dealing with problems with only one equality constraint. We study later problems with two constraints.

We will first recall the notions of tangent line to a level curve and tangent plane to a level surface studied in the chapter on Parametric surfaces II.

0.2 Level curves of differentiable functions

We recall that the level curves of a function $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ are the sets

$$\mathcal{C}_k := \{(x, y) \in \mathbb{R}^2 : g(x, y) = k\} \quad k \in \mathbb{R}.$$

Whenever $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a continuously differentiable function, we want to find the cartesian equation of the tangent line to the level curve $\mathcal{C}_k$ at a point $(a, b) \in \mathcal{C}_k$. To this aim, we will

assume that $\nabla g(a, b) \neq (0, 0)$ and hence, that the piece of the curve $\mathcal{C}_k$ around the point (a, b) is regular and is described by the vector function $\vec{r}: (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}^2$ for some $\varepsilon > 0$ and $\vec{r}(t_0) = (a, b)$ for some $t_0 \in (-\varepsilon, \varepsilon)$.

In that case, we recall that the vector $\vec{r}'(t_0)$ is the direction of the tangent line to the curve $\mathcal{C}_k$ at the point (a, b) .

We get that

$$g(\vec{r}(t)) = k \quad \text{for every } t \in (-\varepsilon, \varepsilon). \quad (1)$$

So, if we differentiate both sides of (1) w.r.t. t we get by the Chain Rule (CR)

$$\nabla g(\vec{r}(t)) \cdot \vec{r}'(t) = 0 \text{ for every } t \in (-\varepsilon, \varepsilon) \quad \underbrace{\Rightarrow}_{t=t_0} \quad \nabla g(\vec{r}(t_0)) \cdot \vec{r}'(t_0) = 0 \Rightarrow \nabla g(a, b) \perp \vec{r}'(t_0).$$

Since the vector $\vec{r}'(t_0)$ is the direction of the tangent line to the curve $\mathcal{C}_k$ at the point (a, b) , it follows that $\nabla g(a, b)$ is perpendicular to that tangent line. Hence, the cartesian equation of that tangent line is given by

$$(x - a, y - b) \cdot \nabla g(a, b) = 0 \quad \Leftrightarrow \quad g_x(a, b)(x - a) + g_y(a, b)(y - b) = 0. \quad (2)$$

0.3 Level surfaces of differentiable functions

We recall that the level surfaces of a function $h: \mathbb{R}^3 \rightarrow \mathbb{R}$ are the sets

$$\mathcal{S}_k := \{(x, y, z) \in \mathbb{R}^3: h(x, y, z) = k\} \quad k \in \mathbb{R}.$$

Whenever, $h: \mathbb{R}^3 \rightarrow \mathbb{R}$ is a continuously differentiable function, we intend to find the cartesian equation of the tangent plane to the level surface $\mathcal{S}_k$ at a point $(a, b, c) \in \mathcal{S}_k$ with $\nabla h(a, b, c) \neq (0, 0, 0)$.

We recall here Definition 0.1 in the chapter on Parametric Surfaces II.

Definition 0.1 (Tangent vector to a surface)

We say that a vector $\vec{u} \in \mathbb{R}^3 \setminus \{(0, 0, 0)\}$ is tangent to the surface $\mathcal{S}_k$ at the point $(a, b, c) \in \mathcal{S}_k$ if there exists a regular curve $\mathcal{C}$ described by a vector function $\vec{r}: (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}^3$ such that:

- (i) $\vec{r}(t) \in \mathcal{S}_k \quad \forall t \in (-\varepsilon, \varepsilon)$;
- (ii) $\vec{r}(t_0) = (a, b, c)$ for some $t_0 \in (-\varepsilon, \varepsilon)$;
- (iii) $\vec{r}'(t_0) = \vec{u}$.

The set of all tangent vector to the surface $\mathcal{S}_k$ at the point (a, b, c) is called the **tangent plane to the surface $\mathcal{S}_k$ at the point (a, b, c)** .

It follows from (i) that

$$h(\vec{r}(t)) = k \quad \forall t \in (-\varepsilon, \varepsilon) \quad \underbrace{\Rightarrow}_{\text{Differentiating (CR)}} \quad \nabla h(\vec{r}(t)) \cdot \vec{r}'(t) = 0 \quad \underbrace{\Rightarrow}_{t=t_0} \quad \nabla h(a, b, c) \cdot \vec{u} = 0. \quad (3)$$

Therefore, $\nabla h(a, b, c)$ is orthogonal to every tangent vector to the level surface $\mathcal{S}_k$ at the point (a, b, c) . So, $\nabla h(a, b, c)$ is orthogonal to the tangent plane to $\mathcal{S}_k$ at the point (a, b, c) and hence, is the normal vector to that plane. Thus, that plane has cartesian equation

$$\begin{aligned} & (x - a, y - b, z - c) \cdot \nabla h(a, b, c) = 0 \\ \Leftrightarrow & \quad h_x(a, b, c)(x - a) + h_y(a, b, c)(y - b) + h_z(a, b, c)(z - c) = 0. \end{aligned} \quad (4)$$

0.4 The method of the Lagrange multipliers

0.4.1 Problems in $\mathbb{R}^2$ with one constraint:

Theorem 0.2 (First theorem on the method of the Lagrange multipliers)

Let $f, g : \mathbb{R}^2 \rightarrow \mathbb{R}$ be functions of class $\mathcal{C}^1$ and let (a, b) be such that

$$g(a, b) = 0 \quad \text{and} \quad \nabla g(a, b) \neq (0, 0).$$

If (a, b) is solution to the problem

$$(P_1) \quad \begin{cases} \text{Find Max/Min of } f(x, y) \\ \text{subject to } g(x, y) = 0. \end{cases}$$

then

$$\nabla f(a, b) // \nabla g(a, b) \Leftrightarrow \exists \lambda \in \mathbb{R} \text{ such that } \nabla f(a, b) = \lambda \nabla g(a, b)$$

$$\Leftrightarrow \exists \lambda \in \mathbb{R} \text{ such that } \begin{cases} f_x(a, b) = \lambda g_x(a, b) \\ f_y(a, b) = \lambda g_y(a, b) \end{cases}$$

Proof: We will assume that (a, b) is a solution of the minimum problem while the maximum problem is treated in a similar way. Since the function g is continuously differentiable $\nabla g(a, b) \neq (0, 0)$, it follows that the level curve $g(x, y) = 0$ around the point (a, b) is regular and is represented by a vector function $\vec{r} : (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}^2$ with $\vec{r}(t_0) = (a, b)$ and also $\nabla g(a, b) \perp \vec{r}'(t_0)$ (*). It follows that $f(\vec{r}(t))$ has a minimum at $t = t_0$ and hence, we get from first year calculus that

$$\frac{d}{dt}(f(\vec{r}(t_0))) = 0 \Leftrightarrow \underbrace{\nabla f(\vec{r}(t_0)) \cdot \vec{r}'(t_0)}_{\text{Chain Rule}} = 0 \Leftrightarrow \nabla f(a, b) \perp \vec{r}'(t_0) (**)$$

Therefore, we obtain from (*) and (**) that $\nabla f(a, b) // \nabla g(a, b)$ which completes the proof of theorem. ■

Remark 0.3 First of all, the number λ whose existence is proved in Theorem 0.6) is called **Lagrange multiplier**.

An easy way to remember the method of the Lagrange multipliers described above is the following: in order to find the solutions (a, b) of the problem (P_1) is to consider the function

$$L(x, y, \lambda) := f(x, y) - \lambda g(x, y)$$

called the **Lagrangian** and find its critical points i.e., find (x, y, λ) such that

$$\nabla L(x, y, \lambda) = (0, 0, 0) \Leftrightarrow \begin{cases} L_x(x, y, \lambda) = 0, \\ L_y(x, y, \lambda) = 0, \\ L_\lambda(x, y, \lambda) = 0 \end{cases}$$

where the last equation $L_\lambda(x, y, \lambda) = 0$ is just the constraint $g(x, y) = 0$.

The method of Lagrange multiplier works for points of maximum as well as minimum. **Therefore, one still needs to evaluate the function f at those points in order to find which ones are points of**

minimum and which ones are points of maximum. However, these points might not be points of **global** maximum/minimum.

In fact, one has to be very careful in the case where the method of the Lagrange multipliers only gives one point. How do you decide whether that point of maximum or minimum?

Example 0.4 Find the points the points of maximum and minimum of the function $f(x, y) := xy$ subject to the constraint $g(x, y) := x^2 + y^2 - 1 = 0$.

Solution: From the description above, we first seek x, y, λ such that

$$\begin{cases} f_x(x, y) = \lambda g_x(x, y), \\ f_y(x, y) = \lambda g_y(x, y) \\ g(x, y) = 0 \end{cases} \Leftrightarrow \begin{cases} y = 2\lambda x, \\ x = 2\lambda y \\ x^2 + y^2 = 1 \end{cases} \quad (*)$$

If we substitute $y = 2\lambda x$ (from the first equation) into the second equation, we get the equation $x(1 - 4\lambda^2) = 0$ which implies that $x = 0$ or $\lambda = \pm \frac{1}{2}$.

However, points (x, y) with $x = 0$ cannot solve the systems of equations (*). In fact, we get from the first equation that also $y = 0$. But $(0, 0)$ does not solve the third equation. So, $x \neq 0$ and hence we necessarily obtain that $\lambda = \pm \frac{1}{2}$.

If $\lambda = -\frac{1}{2}$, then we get from the first two equations that $y = -x$ which substituted into the last equation gives $x = \pm \frac{\sqrt{2}}{2}$. We get the points $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$ and $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$.

One can easily see that if $\lambda = \frac{1}{2}$, we get the points $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$ and $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$.

Now, we evaluate the function f at each of those points and get

$$\boxed{f\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right) = f\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right) = -\frac{1}{2} \quad \text{and} \quad f\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right) = f\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right) = \frac{1}{2}}.$$

Therefore, the points of minimum are

$$\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right) \quad \text{and} \quad \left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$$

while

$$\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right) \quad \text{and} \quad \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right).$$

are the points of maximum.

Exercises 0.5

- Find the points of maximum and minimum for each function subject to the given constraint.

$$(a) \quad f(x, y) := x^3 - 3x(y^2 + 1); \quad g(x, y) := x^2 + y^2 - 1 = 0.$$

$$(b) \quad f(x, y) := x - x^2 - y^2; \quad g(x, y) := x^2 + 2y^2 - 9 = 0.$$

$$(c) \quad f(x, y) := x^3 + y^3; \quad g(x, y) := x^4 + y^4 - 1 = 0.$$

- Find the closest point and the furthest point from the origin $(0, 0)$ on the curve with equation $2x^4 + 2y^4 - xy = 0$.

Problems in $\mathbb{R}^3$ with one constraint:

Theorem 0.6 (Second theorem of the method of the Lagrange multipliers)

Let $f, h : \mathbb{R}^3 \rightarrow \mathbb{R}$ functions of class $\mathcal{C}^1$ and let (a, b, c) be such that $h(a, b, c) = 0$ and $\nabla h(a, b, c) \neq (0, 0, 0)$. If (a, b, c) is solution to the problem

$$(P_2) \quad \begin{cases} \text{Find Max/Min of } f(x, y, z) \\ \text{subject to } h(x, y, z) = 0. \end{cases}$$

then

$$\nabla f(a, b, c) // \nabla h(a, b, c) \Leftrightarrow \exists \lambda \in \mathbb{R} \text{ such that } \nabla f(a, b, c) = \lambda \nabla h(a, b, c)$$

$$\Leftrightarrow \exists \lambda \in \mathbb{R} \text{ such that } \begin{cases} f_x(a, b, c) = \lambda h_x(a, b, c) \\ f_y(a, b, c) = \lambda h_y(a, b, c) \\ f_z(a, b, c) = \lambda h_z(a, b, c) \end{cases}$$

Proof: We assume that (a, b, c) is a point maximum for the function f subject to the constraint $h(x, y, z) = 0$. Since $\nabla h(a, b, c) \neq (0, 0, 0)$, the piece of the level surface $h(x, y, z) = 0$ around the point (a, b, c) is regular and we can assume that it is described by a vector function $\vec{r} : [-\alpha, \alpha] \times [-\beta, \beta] \rightarrow \mathbb{R}^3$ continuously differentiable with $\vec{r}_s(s, t) \times \vec{r}_t(s, t) \neq (0, 0, 0)$ for every $(s, t) \in (-\alpha, \alpha) \times (-\beta, \beta)$. So, each point (x, y, z) on that piece of the level surface $h(x, y, z) = 0$ can be written as

$$(x, y, z) = \vec{r}(s, t) \quad \text{for some } (s, t) \in [-\alpha, \alpha] \times [-\beta, \beta].$$

In particular $(a, b, c) = \vec{r}(s_0, t_0)$ for some $(s_0, t_0) \in [-\alpha, \alpha] \times [-\beta, \beta]$. So, we write

$$h(\vec{r}(s, t)) = 0 \quad \text{for every } (s, t) \in [-\alpha, \alpha] \times [-\beta, \beta].$$

We differentiate w.r.t s and then w.r.t. t and get through the chain rule (CR) that

$$\begin{aligned} \begin{cases} \nabla h(\vec{r}(s, t)) \cdot \vec{r}_s(s, t) = 0, \\ \nabla h(\vec{r}(s, t)) \cdot \vec{r}_t(s, t) = 0, \end{cases} & \xrightarrow{s=s_0, t=t_0} \begin{cases} \nabla h(a, b, c) \cdot \vec{r}_s(s_0, t_0) = 0, \\ \nabla h(a, b, c) \cdot \vec{r}_t(s_0, t_0) = 0. \end{cases} \\ & \Leftrightarrow \boxed{\nabla h(a, b, c) // \vec{r}_s(s_0, t_0) \times \vec{r}_t(s_0, t_0)}. \end{aligned} \quad (5)$$

On the other hand, (a, b, c) is a point of maximum if and only (s_0, t_0) is a point of maximum to the composition function $f \circ \vec{r}(s, t)$. Therefore,

$$\nabla(f \circ \vec{r})(s_0, t_0) = (0, 0) \Leftrightarrow \begin{cases} (f \circ \vec{r})_s(s_0, t_0) = 0, \\ (f \circ \vec{r})_t(s_0, t_0) = 0 \end{cases} \Leftrightarrow \begin{cases} \nabla f(\vec{r}(s_0, t_0)) \cdot \vec{r}_s(s_0, t_0) = 0, \\ \nabla f(\vec{r}(s_0, t_0)) \cdot \vec{r}_t(s_0, t_0) = 0. \end{cases}$$

Hence, we obtain that

$$\boxed{\begin{cases} \nabla f(a, b, c) \cdot \vec{r}_s(s_0, t_0) = 0; \\ \nabla f(a, b, c) \cdot \vec{r}_t(s_0, t_0) = 0 \end{cases}} \Leftrightarrow \nabla f(a, b, c) // \vec{r}_s(s_0, t_0) \times \vec{r}_t(s_0, t_0). \quad (6)$$

Therefore, we finally obtain from (5) and (6) that

$$\boxed{\nabla f(a, b, c) // \nabla h(a, b, c)}$$

and this completes the proof of the theorem. ■

Example 0.7 Find the points of maximum and minimum of the function $f(x, y, z) := y + z$ subject to the constraint $h(x, y, z) := x^2 + y^2 + z^2 - 1 = 0$.

Solution: Following the method of the Lagrange multipliers, we seek x, y, z, λ such that

$$\begin{cases} f_x(x, y, z) = \lambda h_x(x, y, z) \\ f_y(x, y, z) = \lambda h_y(x, y, z) \\ f_z(x, y, z) = \lambda h_z(x, y, z) \\ h(x, y, z) = 0 \end{cases} \Rightarrow \begin{cases} 0 = 2\lambda x \\ 1 = 2\lambda y \\ 1 = 2\lambda z \\ x^2 + y^2 + z^2 = 1 \end{cases}$$

From the second and third equations, we find that $\lambda \neq 0$ and hence, we get that $x = 0$ from the first equation. From the second and third equations we get $y = z = \frac{1}{2\lambda}$ which substituted into the fourth equation give $\lambda = \pm \frac{1}{\sqrt{2}}$. Thus, we get the two points $\left(0, -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$ and $\left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$. So,

$$\begin{cases} f\left(0, -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right) = -\sqrt{2} \Rightarrow \left(0, -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right) \text{ is the point of minimum} \\ f\left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \sqrt{2} \Rightarrow \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \text{ is the point of maximum} \end{cases}$$

Exercises 0.8

- Find the points of maximum and/or minimum for each function subject to the given constraint.
 - $f(x, y, z) := x^2 + y^2 + z^2$ subject to $h(x, y, z) := x + 2y - 3z - 5 = 0$.
 - $f(x, y, z) := e^{x+y^2+z}$ subject to $h(x, y, z) := x^2 + y^2 + z^2 - x - z - 1$.
- Find the parallelepiped with maximum volume and prescribed total area equal to S .
- Find the dimensions of the rectangular box with volume 1000 cm^3 that has minimal surface area.

The next set of notes will be on finding maximum and minimum with multiple equality constraints! Notice that this is not part of the examinable syllabus. Therefore, will not be presented in the lectures!